

# SkillBridge SG

## Cross-Sector Skills Matching for Singapore's Workforce

**"Making every transferable skill count"**

Alibaba Cloud × Qoder Hackathon Singapore 2026

# The Problem

Singapore's SkillsFuture actively encourages **cross-sector career mobility**, yet existing job platforms fail to support it:

- **LinkedIn, MyCareersFuture, JobStreet** use single-sector keyword matching
- A professional accountant with "Data Analytics + Python" is **invisible** to AI Governance roles in ICT — and vice-versa
- Binary skill matching (has / doesn't have) ignores **proficiency levels**
- No lightweight tool understands transferable skills across sectors

# Success Metrics

| Objective                            | Demonstration                        |
|--------------------------------------|--------------------------------------|
| Ranked results in < 1 second         | Live app + latency tests             |
| Proficiency-aware scoring (PL 1 - 6) | matched_detail per result            |
| Cross-sector bridge visible          | Bridge chips + confidence            |
| Every recommendation traceable       | Data provenance                      |
| Inspectable & reproducible           | 97 - test suite + deterministic seed |

# What SkillBridge Does

Paste a Job Description → ranked, explained results in **< 1 second**

- Understands **transferable skills** across sectors
- **Proficiency-aware scoring** (PL 1-6, partial credit)
- Every result is **explainable** (matched, gaps, bridges, courses)
- Accessible to recruiters and career coaches without a data-science team

# The 8-Stage Pipeline (Core Effort)

```
JD → BM25 (top-50) → Dense (top-50) → RRF Fusion (top-30) →  
Cross-Encoder (top-10) → Explainability → Surprise →  
Narrative → Courses
```

**Why hybrid?** Lexical (BM25) catches exact keywords but misses semantic similarity. Dense embeddings catch transferable skills but can drift off-topic.

**Reciprocal Rank Fusion** merges both robustly, and a **cross-encoder** scores each (JD, candidate) pair for precise ranking.

**Funnel:** 150 candidates → 50+50 recalled → 30 fused → 10 re-ranked → scored & explained

# Proficiency-Aware Scoring (Differentiator)

Official SWDA proficiency scale:

**1 Use → 2 Operate → 3 Apply → 4 Implement → 5 Analyse → 6 Assess/Lead**

**Per-skill credit:**

- Full credit `1.0` if `candidate_PL ≥ required_PL`
- Partial credit `candidate_PL / required_PL` if below required level
- Zero if skill is absent

**Hybrid display score:** `40 + 58 × (0.3 × semantic_norm + 0.7 × proficiency_fit)` → range 40-98

70% weight on proficiency fit ensures **real skill match dominates text similarity**

# Bridge Skills (Transferable Evidence)

A bridge answers: *"Candidate lacks skill X, but has skill Y — how much of X is covered by Y?"*

**Derived from 150K+ Knowledge & Ability (K&A) items** in the official SWDA dataset

Containment score:  $|S \cap V| / |S|$ , floor 0.2, Critical Core Skills (CCS) +0.1 boost

**53 evidence-based bridges** with explicit confidence scores (0.20-0.87)

# Data Provenance (Not a Black Box)

All data grounded in **official SWDA Skills Framework dataset**:

| Seed File           | Count        | Key Details                                                  |
|---------------------|--------------|--------------------------------------------------------------|
| jobs.json           | 30 roles     | 6 × 5 sectors, skill_requirements verbatim                   |
| candidates.json     | 150 profiles | Anchored to real roles, ~78% skill coverage                  |
| skill_registry.json | 434 skills   | TSC/CCS titles, Emerging/CASL flags                          |
| skill_taxonomy.json | 5 sectors    | finance 70, ict 63, healthcare 80, eng 64, sustainability 77 |
| bridges.json        | 53 bridges   | K&A containment, floor 0.2, CCS +0.1 boost                   |
| courses.json        | 434 entries  | Live MySkillsFuture search URLs                              |

**Deterministic:** `random.seed(20260729)` → byte-identical output, auditable, validated



# What It Can Do

- ✓ Match JD → ranked results in < 1 second
- ✓ Proficiency-aware scoring (PL 1-6) grounded in official data
- ✓ Cross-sector bridge skills with evidence-backed confidence
- ✓ Explainability panel: matched, gaps, bridges, courses
- ✓ What-if explorer: add/remove skills, see scores shift ▲/▼
- ✓ Hire ↔ Upskill perspective toggle
- ✓ Surprise mode: surface serendipitous candidates
- ✓ Emerging & CASL badges from official dataset
- ✓ Sector convergence strip (Jaccard overlap visualization)
- ✓ 97 tests, fully reproducible, deterministic

**Cannot:** production-scale talent pool, real resume parsing, open-ended NLP skill extraction

# Demo

**Video:** <https://youtu.be/DhmpkfX31yg>

**GitHub:** [https://github.com/L3sli3Ch1ang/qoder\\_hackathon.git](https://github.com/L3sli3Ch1ang/qoder_hackathon.git)

Live app: FastAPI + Alpine.js + Tailwind

Run locally:

```
uvicorn app.main:app --host 127.0.0.1 --port 8000
```

# Built with Qoder

**Quest Mode:** spec + scaffold, SWDA data integration, 6-phase development

**Expert Mode:** 8-stage pipeline, proficiency scoring, bridge derivation from 150K+ K&A items

**qodercli:** test runs, isolated worktree, `/review`, `/init`

**2 weeks**, 97 tests, real government data, deterministic seed generation

Evidence: `AGENTS.md`, `docs/evidence/`, `docs/evidence/code-reviews/`

**Stack:** FastAPI + Alpine.js + Tailwind, sentence-transformers, Qdrant Lite, DashScope qwen-plus, NixOS + direnv

# Roadmap

## Future investments (in priority order):

1. **Real candidate ingestion** — parse PDF/DOCX resumes, map to SWDA taxonomy
2. **Scale corpus** — expand from 5 to 38 SWDA sectors (2,030 roles / 12,007 skills)
3. **Richer skill extraction** — NER/embedding-based detection beyond taxonomy
4. **Live data sync** — periodic re-derivation from SWDA/MySkillsFuture releases
5. **Persistence & accounts** — save searches, track pipelines, multi-user
6. **Outcome feedback loop** — tune blend weights based on actual hires
7. **Bias & fairness review** — audit scorer for sector/seniority bias

# Thank You

Questions?

**Demo:** <https://youtu.be/DhmpkfX31yg>

**GitHub:** [https://github.com/L3sli3Ch1ang/qoder\\_hackathon.git](https://github.com/L3sli3Ch1ang/qoder_hackathon.git)

#QoderHackathon #BuildWithQoder #SkillsFuture #AlibabaCloud